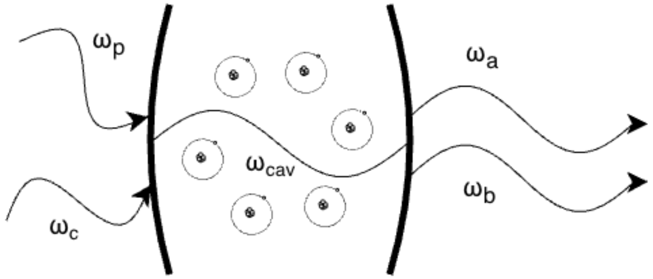
Simulation of Squeezed Light Generation in an Optical Resonator

Student: Aaron Dayton – Supervisor: Prof. Andrew MacRae

Project Idea



Spherical resonator containing many 3-level (Λ-system) atoms

Master Equation

$$\frac{d\rho}{\mathrm{d}t} = -\frac{i}{\hbar}[H,\rho] + \sum_i \gamma_i \left(L_i \rho L_i^\dagger - \frac{1}{2} L_i^\dagger L_i \rho - \frac{1}{2} \rho L_i^\dagger L_i \right)$$

Four Wave Mixing

- Optical nonlinearities

$$P = \varepsilon_0 (\chi^{(1)} E + \chi^{(2)} E^2 + \chi^{(3)} E^3 + \cdots)$$

- Produces a squeezed state

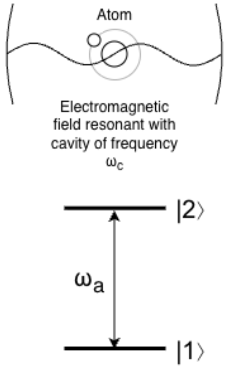
$$\Delta x \, \Delta p \geq \frac{\hbar}{2}$$

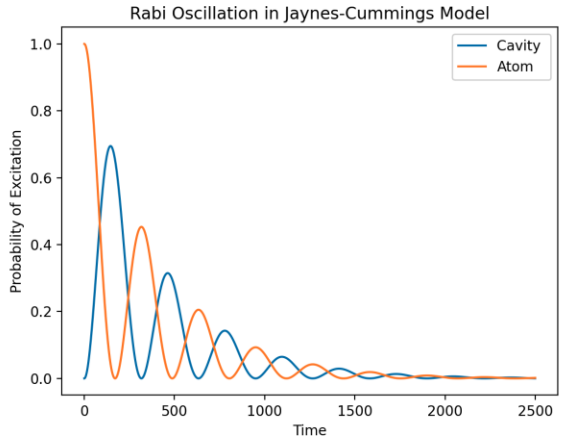
- Frequency conversion

Electromagnetically Induced Transparency

- Narrow transparency window for frequencies of light
- Using a cavity narrows transparency window even further

Jaynes-Cummings Model





Electromagnetically Induced Transparency with a 3-level Atom